Verifier-guided inference-time search for language models

A. Example and B. Research

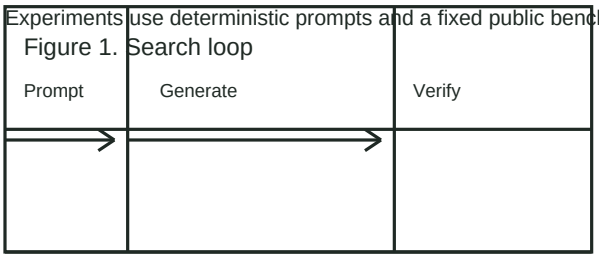
Abstract

We study verifier-guided search for language-model inference.
The method evaluates candidate trajectories against a formal verifier.

1 Introduction

Inference-time search can improve model reliability without retraining.
We compare a direct decoder with a verifier-guided search procedure.
Each candidate trajectory is checked against the specification.
The search retains the highest scoring valid path at each step.
The proof obligation is represented as a sequence of local constraints.
Experiments use deterministic prompts and a fixed public benchmark.

Figure 1. Search loop



The diagram is rendered from vector source paths.

2 Method

The verifier computes a Boolean judgment for every proposed step.
Invalid transitions are discarded before the next search expansion.
The beam is ordered by model likelihood and verifier acceptance.
Ablations remove either the verifier or the look-ahead operation.
All reported values are measured from the source evaluation runs.

Table 1. Benchmark results

Method	Pass	Steps
Direct	0.71	8.2
Verified	0.88	6.4

3 Search objective

$$g_t = \arg\max_g P(y \mid x, g) \text{ subject to } V(g) = 1$$

Equation 1. Select a valid trajectory with maximum conditional likelihood.

References

[1] A. Example, Search with verifiers, 2025. doi:10.1234/example.1

[2] B. Research, Formal inference methods, 2024. arXiv:2401.12345